

Homework 14

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Problem 14.1.

Solution 14.1.

1.

$$\begin{bmatrix} MN & O \\ N & O \end{bmatrix} \xrightarrow[\text{row operation}]{\begin{bmatrix} I_m & -M \\ & I_n \end{bmatrix}} \begin{bmatrix} O & O \\ N & O \end{bmatrix} \xrightarrow[\text{column operation}]{\begin{bmatrix} I_m & M \\ & I_n \end{bmatrix}} \begin{bmatrix} O & O \\ N & NM \end{bmatrix}$$

So when $X = \begin{bmatrix} I_m & M \\ & I_n \end{bmatrix}$, we have $X^{-1}AX = B \implies A = XBX^{-1} \implies X = \begin{bmatrix} I_m & M \\ & I_n \end{bmatrix}$.

2.

$$A = X_1DX_1^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$B = X_2DX_2^{-1} = \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{bmatrix}$$

Then

$$A = X_1DX_1^{-1} = X_1(X_2^{-1}BX_2)X_1^{-1} = (X_1X_2^{-1})B(X_1X_2^{-1})^{-1}$$

So $X = X_1X_2^{-1} = \frac{1}{2} \begin{bmatrix} 1-i & 1-i \\ 1+i & -1-i \end{bmatrix}$.

3.

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ & 1 & 3 \\ & & 1 \end{bmatrix} &\xrightarrow[\text{row operation}]{\begin{bmatrix} 1 & -1 & \\ & 1 & \\ & & 1 \end{bmatrix}} \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \xrightarrow[\text{column operation}]{\begin{bmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{bmatrix}} \begin{bmatrix} 1 & 2 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 2 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} &\xrightarrow[\text{row operation}]{\begin{bmatrix} \frac{1}{2} & & \\ & 1 & \\ & & 1 \end{bmatrix}} \begin{bmatrix} \frac{1}{2} & 1 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \xrightarrow[\text{column operation}]{\begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix}} \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 3 \\ & & 1 \end{bmatrix} &\xrightarrow[\text{row operation}]{\begin{bmatrix} 1 & & \\ & 1 & \\ & & 3 \end{bmatrix}} \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 3 \\ & & 3 \end{bmatrix} \xrightarrow[\text{column operation}]{\begin{bmatrix} 1 & & \\ & 1 & \\ & & \frac{1}{3} \end{bmatrix}} \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \end{aligned}$$

Hence,

$$\begin{bmatrix} 1 & 1 & 0 \\ & 1 & 1 \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & & \frac{1}{3} \end{bmatrix}$$

Note that the column operations and row operations are inverse of each others. Then,

$$B = \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 1 \\ & & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & \\ & 1 & \\ & & \frac{1}{3} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 3 \\ & 1 & 3 \\ & & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & \\ & 1 & \\ & & \frac{1}{3} \end{bmatrix} = X^{-1}AX$$

So $A = XBX^{-1}$ and $X = \begin{bmatrix} 2 & 1 & \\ & 1 & \\ & & \frac{1}{3} \end{bmatrix}$.

4. Diagonalization

$$\begin{bmatrix} 4 & -2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}^{-1} \implies X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$$

5. Similar to subproblem 4, $X = \begin{bmatrix} I & 2I \\ I & 3I \end{bmatrix}$.

6. Note that

$$\begin{bmatrix} m & -n \\ n & m \end{bmatrix} = \begin{bmatrix} i & 1 \\ 1 & i \end{bmatrix} \begin{bmatrix} m + in & \\ & m - in \end{bmatrix} \begin{bmatrix} i & 1 \\ 1 & i \end{bmatrix}^{-1}$$

Therefore,

$$\begin{bmatrix} M & -N \\ N & M \end{bmatrix} = \begin{bmatrix} iI & I \\ I & iI \end{bmatrix} \begin{bmatrix} M + iN & \\ & M - Ni \end{bmatrix} \begin{bmatrix} iI & I \\ I & iI \end{bmatrix}^{-1}$$

So $X = \begin{bmatrix} iI & I \\ I & iI \end{bmatrix}$.

7. Note that

$$\begin{bmatrix} 2 & m \\ & 3 \end{bmatrix} = \begin{bmatrix} 1 & m \\ & 1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 3 \end{bmatrix} \begin{bmatrix} 1 & -m \\ & 1 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 2I_2 & M \\ & 3I_3 \end{bmatrix} = \begin{bmatrix} I_2 & M \\ & I_3 \end{bmatrix} \begin{bmatrix} 2I_2 & \\ & 3I_3 \end{bmatrix} \begin{bmatrix} I_2 & -M \\ & I_3 \end{bmatrix}$$

So $X = \begin{bmatrix} I_2 & M \\ & I_3 \end{bmatrix}$.

Problem 14.2.

Solution 14.2.

1. Set $H_n = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_n]$, where $\mathbf{q}_1, \dots, \mathbf{q}_n$ are mutually perpendicular and made of ± 1 . Also note that H_n is a symmetric matrix, then

$$H_n^2 = H_n^T H_n = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_n] = \begin{bmatrix} \mathbf{q}_1^T \mathbf{q}_1 & \mathbf{q}_1^T \mathbf{q}_2 & \cdots & \mathbf{q}_1^T \mathbf{q}_n \\ \mathbf{q}_2^T \mathbf{q}_1 & \mathbf{q}_2^T \mathbf{q}_2 & \cdots & \mathbf{q}_2^T \mathbf{q}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_n^T \mathbf{q}_1 & \mathbf{q}_n^T \mathbf{q}_2 & \cdots & \mathbf{q}_n^T \mathbf{q}_n \end{bmatrix} = \begin{bmatrix} n & & & \\ & n & & \\ & & \ddots & \\ & & & n \end{bmatrix} = nI$$

Since $H_n^2 = nI$, and the eigenvalues of nI are n , by HW 13.1, the eigenvalues of H_n are $\pm\sqrt{n}$.

2. If n is some power of 2 and $n > 1$, then $H_n = \begin{bmatrix} H_{\frac{n}{2}} & H_{\frac{n}{2}} \\ H_{\frac{n}{2}} & -H_{\frac{n}{2}} \end{bmatrix}$. So $\text{trace}(H_n) = \text{trace}(H_{\frac{n}{2}}) + \text{trace}(-H_{\frac{n}{2}}) = 0$.

Thus,

$$M_a(\sqrt{n}) = M_a(-\sqrt{n}) = \frac{n}{2}.$$

3.

$$\begin{aligned} \frac{1}{\sqrt{2}} H_2 \mathbf{e}_1 &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ \frac{1}{\sqrt{2}} H_2 \mathbf{e}_2 &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \end{aligned}$$

So it's a reflection about the line with slope angle $\frac{\pi}{8}$. Obviously, this reflection fixes every vectors in this line.

$$\frac{1}{\sqrt{2}} H_2 \begin{bmatrix} \cos \frac{\pi}{8} \\ \sin \frac{\pi}{8} \end{bmatrix} = \begin{bmatrix} \cos \frac{\pi}{8} \\ \sin \frac{\pi}{8} \end{bmatrix}$$

And the reflection also maps all the vectors that are perpendicular to the line to its reverse direction.

$$\frac{1}{\sqrt{2}} H_2 \begin{bmatrix} \cos \frac{5\pi}{8} \\ \sin \frac{5\pi}{8} \end{bmatrix} = - \begin{bmatrix} \cos \frac{5\pi}{8} \\ \sin \frac{5\pi}{8} \end{bmatrix}$$

So the eigenvalues of $\frac{1}{\sqrt{2}} H_2$ are $\lambda_1 = 1, \lambda_2 = -1$, and the corresponding eigenvectors are $\text{span}\left(\begin{bmatrix} \cos \frac{\pi}{8} \\ \sin \frac{\pi}{8} \end{bmatrix}\right)$ and

$\text{span}\left(\begin{bmatrix} \cos \frac{5\pi}{8} \\ \sin \frac{5\pi}{8} \end{bmatrix}\right)$, respectively.

4.

Method 1:

Similar to Homework 14.1.6, we consider the 1×1 case first. Set $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, the eigenvalues are $\lambda_1 = \sqrt{2}, \lambda_2 = -\sqrt{2}$, and the eigenvectors are $\text{span}\left(\begin{bmatrix} 1+\sqrt{2} \\ 1 \end{bmatrix}\right), \text{span}\left(\begin{bmatrix} 1-\sqrt{2} \\ 1 \end{bmatrix}\right)$, respectively. Then the eigenvectors of $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$ for eigenvalues $\sqrt{2}\lambda$ is $\text{span}\left(\begin{bmatrix} (1+\sqrt{2})\mathbf{v} \\ \mathbf{v} \end{bmatrix}\right)$, the eigenvectors of $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$ for eigenvalues $-\sqrt{2}\lambda$ is $\text{span}\left(\begin{bmatrix} (1-\sqrt{2})\mathbf{v} \\ \mathbf{v} \end{bmatrix}\right)$.

Method 2:

Set an eigenvector of H_{2n} for the eigenvalue $\sqrt{2}\lambda$ is $\begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix}$, then

$$H_{2n} \begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix} \begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix} = \begin{bmatrix} (s+t)H_n\mathbf{v} \\ (s-t)H_n\mathbf{v} \end{bmatrix} = \begin{bmatrix} (s+t)\lambda\mathbf{v} \\ (s-t)\lambda\mathbf{v} \end{bmatrix}$$

By the definition of eigenvectors and eigenvalues, we have $\begin{cases} (s+t)\lambda = \sqrt{2}s\lambda \\ (s-t)\lambda = \sqrt{2}t\lambda \end{cases} \implies s = (1+\sqrt{2})t$. So the eigenvectors for $\sqrt{2}\lambda$ is $\begin{bmatrix} (1+\sqrt{2})t\mathbf{v} \\ t\mathbf{v} \end{bmatrix}, t \in \mathbb{C}$, which is equivalent to $\text{span}\left(\begin{bmatrix} (1+\sqrt{2})\mathbf{v} \\ \mathbf{v} \end{bmatrix}\right)$.

Similarly, set an eigenvector of H_{2n} for the eigenvalue $-\sqrt{2}\lambda$ is $\begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix}$, then

$$H_{2n} \begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix} \begin{bmatrix} s\mathbf{v} \\ t\mathbf{v} \end{bmatrix} = \begin{bmatrix} (s+t)H_n\mathbf{v} \\ (s-t)H_n\mathbf{v} \end{bmatrix} = \begin{bmatrix} (s+t)\lambda\mathbf{v} \\ (s-t)\lambda\mathbf{v} \end{bmatrix}$$

By the definition of eigenvectors and eigenvalues, we have $\begin{cases} (s+t)\lambda = -\sqrt{2}s\lambda \\ (s-t)\lambda = -\sqrt{2}t\lambda \end{cases} \implies s = (1-\sqrt{2})t$. So the eigenvectors for $-\sqrt{2}\lambda$ is $\text{span}\left(\begin{bmatrix} (1-\sqrt{2})\mathbf{v} \\ \mathbf{v} \end{bmatrix}\right)$.

Problem 14.3.

Solution 14.3.

1. By observation, $\det(A) = -8, \text{tr}(A) = 3$. So the eigenvalues are $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = -2$. The corresponding eigenvectors are $\text{span}\left(\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}\right), \text{span}\left(\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}\right), \text{span}\left(\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}\right)$, respectively. For real symmetric matrices, eigenvectors from different eigenvalues are mutually perpendicular. So we only need to divide their length to make them ONB. Then the Orthonormal Basis is $\frac{1}{3}\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \frac{1}{3}\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. And the Spectral Decomposition is

$$A = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 4 & & \\ & 1 & \\ & & -2 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} \frac{1}{3}$$

2. By observation, $A - I = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{bmatrix}$ is not invertible, so $\lambda = 1$ is one of the eigenvalues. And $\text{rank}(A - I) = 1 \implies M_g(1) = 2 \implies M_a(1) \geq 2$. But $\text{trace}(A) = 12$, so we have $1 + 1 + \lambda_3 = 12 \implies \lambda_3 = 10$. Hence the eigenvalues are $\lambda_1 = \lambda_2 = 1, \lambda_3 = 10$.

Alternatively, you can also make use of $\text{trace}(A) = 12, \det(A) = 10$ to get $\lambda_1 = \lambda_2 = 1, \lambda_3 = 10$.

The eigenvectors for $\lambda = 1$ are $\text{span}\left(\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right)$. The eigenvectors for $\lambda = 10$ are $\text{span}\left(\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}\right)$. Note that $\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are not mutually perpendicular because they are from the same eigenvalues. So we need to do Gram-Schmidt

Orthogonalization to make them ONB. Then the Orthonormal Basis is $\begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{2}{3\sqrt{5}} \\ \frac{4}{3\sqrt{5}} \\ \frac{5}{3\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}$. And the Spectral

Decomposition is

$$A = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} & \frac{1}{3} \\ -\frac{1}{\sqrt{5}} & \frac{3\sqrt{5}}{4} & \frac{2}{3} \\ 0 & \frac{5}{3\sqrt{5}} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & & 10 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{4\sqrt{5}} & 0 \\ \frac{1}{3\sqrt{5}} & \frac{2}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

3. Note that the Spectral Decomposition of $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ is $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}}$.

Set the Spectral Decomposition of A is $A = Q\Lambda Q^T$, then the Spectral Decomposition of $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$ is

$$\begin{bmatrix} A & A \\ A & A \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} Q & Q \\ Q & -Q \end{bmatrix} \begin{bmatrix} \Lambda & \\ & -\Lambda \end{bmatrix} \begin{bmatrix} Q^T & Q^T \\ Q^T & -Q^T \end{bmatrix} \frac{1}{\sqrt{2}}$$

Set $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$, then spectral decomposition $A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & \\ & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} = Q\Lambda Q^T$.

So the Spectral Decomposition is

$$\begin{aligned} \begin{bmatrix} 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 4 \\ 4 & 1 & 0 & 0 \\ 1 & 4 & 0 & 0 \end{bmatrix} &= \begin{bmatrix} A & A \\ A & A \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} Q & Q \\ Q & -Q \end{bmatrix} \begin{bmatrix} \Lambda & \\ & -\Lambda \end{bmatrix} \begin{bmatrix} Q^T & Q^T \\ Q^T & -Q^T \end{bmatrix} \frac{1}{\sqrt{2}} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 5 & & & \\ & 3 & & \\ & & -5 & \\ & & & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \frac{1}{2} \end{aligned}$$

4. Similar to subproblem 3, we have $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}}$.

Set $A = Q\Lambda Q^T$, then $\begin{bmatrix} A & A \\ A & A \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} Q & Q \\ Q & -Q \end{bmatrix} \begin{bmatrix} 2\Lambda & \\ & O \end{bmatrix} \begin{bmatrix} Q^T & Q^T \\ Q^T & -Q^T \end{bmatrix} \frac{1}{\sqrt{2}}$. So the Spectral Decomposition is

$$\begin{aligned} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} &= \begin{bmatrix} A & A \\ A & A \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} Q & Q \\ Q & -Q \end{bmatrix} \begin{bmatrix} 2\Lambda & \\ & O \end{bmatrix} \begin{bmatrix} Q^T & Q^T \\ Q^T & -Q^T \end{bmatrix} \frac{1}{\sqrt{2}} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \frac{1}{2} \end{aligned}$$

5. A has 3 eigenvalues, so it's a 3×3 matrix. Since $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvectors for 0, and eigenvectors from different

eigenvalues are mutually perpendicular, then the two eigenvectors for 3 are $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. After GS orthogonalization, the spectral decomposition is

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 3 & & \\ & 3 & \\ & & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}^T$$

6. $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \implies$ one of eigenvalues is 3, and the eigenvectors are $\text{span}(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix})$.

$$\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \in \ker(A) \implies \text{two eigenvalues are 0, and the corresponding eigenvectors are } \text{span}\left(\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}\right).$$

After GS orthogonalization, the spectral decomposition is

$$A = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 3 & & \\ & 0 & \\ & & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}^T$$

7. By observation, $a + 1$ is one of the eigenvalues, because $A - (a + 1)I = \begin{bmatrix} -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$ is not invertible.

And $\text{rank}(A - (a + 1)I) = 1$, so $M_g(a + 1) = \dim \ker(A - (a + 1)I) = 3 \implies M_a(a + 1) \geq 3$. But we have a simple (single) eigenvalue $-3 \implies M_a(a + 1) = 3, M_a(-3) = 1$. Therefore,

$$\text{trace}(A) = 4a = 3(a + 1) + (-3) \implies a = 0$$

So the eigenvalues of A are $1, 1, 1, -3$. The spectral decomposition is

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}^T$$

8. $A^3 = O \implies \lambda^3 = 0 \implies \lambda_1 = \lambda_2 = \lambda_3 = 0 \implies A = QOQ^T$, where Q is any orthogonal matrix.

9. $A^5 = I \implies \lambda^5 = 1 \implies \lambda = 1, M_a(1) = 5 \implies A = QIQ^T$, where Q is any orthogonal matrix.

10.

$$A = \begin{bmatrix} \mathbf{u} & \mathbf{v} \end{bmatrix} \begin{bmatrix} a & \\ & b \end{bmatrix} \begin{bmatrix} \mathbf{u}^T \\ \mathbf{v}^T \end{bmatrix}$$

Problem 14.4.

Solution 14.4.

1.
 $\det(A) = -b \implies A$ is invertible if and only if $b \neq 0$.
 The characteristic polynomial of A is $p_A(x) = x^2 - 2x - b$. When $(-2)^2 - 4(-b) = 4 + 4b \neq 0 \implies b \neq -1$, A has two distinct eigenvalues, it is diagonalizable. When $(-2)^2 - 4(-b) = 4 + 4b = 0 \implies b = -1$, $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$, the eigenvalues $\lambda_1 = \lambda_2 = 1$, but $M_g(1) = 1$, so it's not diagonalizable.
 If $b = 1$, then A is real symmetric, it can be diagonalized by orthogonal matrices.

2.
 $A^2 = O$. $A^* \neq A \implies$ not Hermitian. $A^* \neq -A \implies$ not skew-Hermitian. $A^*A = \begin{bmatrix} 2 & -2i \\ 2i & 2 \end{bmatrix} \neq I_2 \implies$ not Unitary. And $AA^* = \begin{bmatrix} 2 & 2i \\ -2i & 2 \end{bmatrix} \neq A^*A \implies$ not Normal. Finally, $\text{trace}(A) = 0$, but $\det(A) = 0$, so $\lambda_1 = \lambda_2 = 0$, but $M_g(0) = 1 \implies$ not diagonalizable.

Problem 14.5.

Solution 14.5.

1. Note that we only have two eigenvalues λ_1, λ_2 , then λ_1 and λ_2 are all positive if and only if $\lambda_1 + \lambda_2 = \text{trace}(A) > 0$ and $\lambda_1\lambda_2 = \det(A) > 0$, they are all negative if and only if $\lambda_1 + \lambda_2 = \text{trace}(A) < 0$ and $\lambda_1\lambda_2 = \det(A) > 0$, they are indefinite if and only if $\lambda_1\lambda_2 = \det(A) < 0$. And we have $\text{trace}(A) = 2c, \det(A) = c^2 - b^2$.

$$\text{Positive Definite} \iff \text{trace}(A) > 0, \det(A) > 0 \iff \begin{cases} c > 0 \\ -c < b < c \end{cases}.$$

$$\text{Positive SemiDefinite} \iff \text{trace}(A) \geq 0, \det(A) \geq 0 \iff \begin{cases} c \geq 0 \\ -c \leq b \leq c \end{cases}.$$

$$\text{Negative definite} \iff \text{trace}(A) < 0, \det(A) > 0 \iff \begin{cases} c < 0 \\ c < b < -c \end{cases}.$$

$$\text{Negative Semidefinite} \iff \text{trace}(A) \leq 0, \det(A) \geq 0 \iff \begin{cases} c \leq 0 \\ c \leq b \leq -c \end{cases}.$$

Indefinite $\iff \det(A) < 0 \iff b > |c|$ or $b < -|c|$.

2. $\text{trace}(A) = 10 > 0$, so it's impossible to be Negative definite or Negative Semidefinite.

Positive Definite $\iff \det(A) > 0 \iff -3 < b < 3$.

Similarly, Positive SemiDefinite $\iff \det(A) \geq 0 \iff -3 \leq b \leq 3$.

Indefinite $\iff b > 3$ or $b < -3$.

3. $\text{trace}(A) = 2 + c, \det(A) = 2c - 16$.

Positive Definite $\iff c > 8$.

Positive SemiDefinite $\iff c \geq 8$.

Negative definite and Negative Semidefinite are impossible.

Indefinite $\iff c < 8$.

4. Set $X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then $A = \begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix} = X + (c-1)I$. The eigenvalues of X are $0, 0, 3 \implies$ eigenvalues of A are $c-1, c-1, c+2$.

Positive Definite $\iff c > 1$.

Positive SemiDefinite $\iff c \geq 1$.

Negative definite $\iff c < -2$.

Negative Semidefinite $\iff c \leq -2$.

Indefinite $\iff -2 < c < 1$.

5. Similar to subproblem 4, the eigenvalues of A are $c+4, c+4, c-8$.

Positive Definite $\iff c > 8$.

Positive SemiDefinite $\iff c \geq 8$.

Negative definite $\iff c < -4$.

Negative Semidefinite $\iff c \leq -4$.

Indefinite $\iff -4 < c < 8$.

6. By Gaussian Elimination, and assume that $d \neq 4$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & d-4 & -2 \\ 0 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & d-4 & -2 \\ 0 & 0 & -4 - \frac{4}{d-4} \end{bmatrix}$$

When $d = 4$, A is indefinite.

$$A \text{ is Positive Definite} \iff \begin{cases} 1 > 0 \\ d-4 > 0 \\ -4 - \frac{4}{d-4} > 0 \end{cases} \implies \text{no solution.}$$

Similarly, Positive SemiDefinite, Negative definite, and Negative Semidefinite also impossible.

So when $d \in \mathbb{R}$, A is indefinite.

7. Set $X = \begin{bmatrix} 0 & 3 & 0 \\ 3 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix}$, then $A = X + tI$. The eigenvalues of X are $0, 5, -5$, so the eigenvalues of A are $t, 5+t, -5+t$.

Positive Definite $\iff t > 5$.

Positive SemiDefinite $\iff t \geq 5$.

Negative definite $\iff t < -5$.

Negative Semidefinite $\iff t \leq -5$.

Indefinite $\iff -5 < t < 5$.

8.

Method 1:

By Gaussian Elimination,

$$\begin{bmatrix} a & a & a \\ a & a+c & a-c \\ a & a-c & a+c \end{bmatrix} \rightarrow \begin{bmatrix} a & a & a \\ 0 & c & -c \\ 0 & -c & c \end{bmatrix} \rightarrow \begin{bmatrix} a & a & a \\ 0 & c & -c \\ 0 & 0 & 0 \end{bmatrix}$$

So the matrix is not invertible, Positive Definiteness and Negative Definiteness are all impossible.

Positive SemiDefinite $\iff a > 0, c > 0$.

Negative SemiDefinite $\iff a < 0, c < 0$.

Indefinite $\iff ac < 0$.

Method 2:

By observation, $\begin{bmatrix} a & a & a \\ a & a+c & a-c \\ a & a-c & a+c \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3a \\ 3a \\ 3a \end{bmatrix} = 3a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. So one of the eigenvalues is $\lambda_1 = 3a$.

$$\det(A) = \begin{vmatrix} a & a & a \\ a & a+c & a-c \\ a & a-c & a+c \end{vmatrix} = \begin{vmatrix} a & a & a \\ 0 & c & -c \\ 0 & -c & c \end{vmatrix} = 0, \text{ so one of the eigenvalues is } \lambda_2 = 0.$$

$$\text{trace}(A) = 3a + 2c \implies \lambda_3 = 2c.$$

Hence the eigenvalues of A are $3a, 0, 2c$.

Positive Definiteness and Negative Definiteness are all impossible.

Positive SemiDefinite $\iff a > 0, c > 0$.

Negative SemiDefinite $\iff a < 0, c < 0$.

Indefinite $\iff ac < 0$.

Problem 14.6.

Solution 14.6.

1. Proof:

$A - B$ is Positive SemiDefinite, $B - C$ is Positive SemiDefinite $\implies \forall \mathbf{x} \in \mathbb{C}^n, \mathbf{x}^*(A - B)\mathbf{x} \geq 0, \mathbf{x}^*(B - C)\mathbf{x} \geq 0$.

Thus, $\forall \mathbf{x} \in \mathbb{C}^n$,

$$\mathbf{x}^*(A - C)\mathbf{x} = \mathbf{x}^*(A - B + B - C)\mathbf{x} = \mathbf{x}^*(A - B)\mathbf{x} + \mathbf{x}^*(B - C)\mathbf{x} \geq 0$$

So $A - C$ is Positive SemiDefinite.

2. Proof:

Note that $B - A = -(A - B)$, and $B - A, A - B$ are both Hermitian. Set the eigenvalues of $A - B$ are $\lambda_1, \dots, \lambda_n$, then the eigenvalues of $B - A$ are $-\lambda_1, \dots, -\lambda_n$.

By positive semidefiniteness of $A - B$ and $B - A$, $\forall 1 \leq i \leq n, \begin{cases} \lambda_i \geq 0 \\ -\lambda_i \geq 0 \end{cases} \implies \lambda_i = 0 \implies$ all the eigenvalues of

$A - B$ are 0.

Since $A - B$ are Hermitian, it can be spectral decomposition, that is,

$$A - B = QOQ^T = O \implies A = B$$

3. Proof:

Recall that the eigenvalues of Hermitian matrices are all real number. Set the eigenvalues of A are $\lambda_1, \dots, \lambda_n$, then eigenvalues of $A - aI$ are $\lambda_1 - a, \dots, \lambda_n - a$. To make $A - aI$ Positive SemiDefinite, we only need $a \leq \lambda_{\min}$. Similarly, to make $bI - A$ Positive SemiDefinite, we only need $b \geq \lambda_{\max}$. So for any Hermitian A , we can find $a, b \in \mathbb{R}$ such that $A - aI$ and $bI - A$ are Positive SemiDefinite.

Problem 14.7.

Solution 14.7.

1. True.

Proof:

Set the eigenvalues of A are $\lambda_1, \lambda_2, \dots, \lambda_n$, then the eigenvalues of A^{-1} are $\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}$.

A is positive definite $\implies \lambda_1, \lambda_2, \dots, \lambda_n > 0 \implies \lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1} > 0 \implies A^{-1}$ is positive definite.

2. True.

Proof:

A, B are positive definite $\implies \forall \mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}, \mathbf{x}^*A\mathbf{x} > 0, \mathbf{x}^*B\mathbf{x} > 0$. Therefore, $\forall \mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}$

$$\mathbf{x}^*(A + B)\mathbf{x} = \mathbf{x}^*A\mathbf{x} + \mathbf{x}^*B\mathbf{x} > 0$$

So $A + B$ is positive definite.

3. True.

Proof similar to subproblem 2.

4. False.

Set $A = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}, B = \begin{bmatrix} -1 & \\ & 2 \end{bmatrix}$. They are all indefinite but $A + B = \begin{bmatrix} 0 & \\ & 1 \end{bmatrix}$ is positive semidefinite.

5. True.

Proof:

If A has full column rank, then $\forall \mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}, A\mathbf{x} \neq \mathbf{0}$. Therefore, $\forall \mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}$, we always have

$$\mathbf{x}^*A^*BA\mathbf{x} = (A\mathbf{x})^*B(A\mathbf{x}) > 0$$

So A^*BA is positive definite.

6. True.

Proof:

Set λ is an arbitrary eigenvalue of AB , one of the corresponding eigenvectors is $\mathbf{x}$. Then

$$AB\mathbf{x} = \lambda\mathbf{x}$$

Left multiplying $\mathbf{x}^* B^*$ simultaneously,

$$\mathbf{x}^* B^* A B \mathbf{x} = \lambda \mathbf{x}^* B^* \mathbf{x} \implies (B \mathbf{x})^* A (B \mathbf{x}) = \lambda \mathbf{x}^* B^* \mathbf{x}$$

Since A is positive definite, $(B \mathbf{x})^* A (B \mathbf{x}) > 0$. At the same time, $\mathbf{x}^* B^* \mathbf{x} = (\mathbf{x}^* B^* \mathbf{x})^* = \mathbf{x}^* B \mathbf{x} > 0$. Hence $\lambda > 0$.

7. True.

Proof similar to subproblem 1.

8. True.

Proof:

Let's prove a lemma first.

Lemma 1: If $x_1, x_2, \dots, x_s$ are all positive and distinct, then there exists a degree $s - 1$ polynomial $p(x)$ such that $p(x_1^r) = x_1, p(x_2^r) = x_2, \dots, p(x_s^r) = x_s$.

Proof:

Set $y_1 = x_1^r, \dots, y_n = x_n^r$, then $y_1, y_2, \dots, y_n$ are all distinct, and $p(y_1) = x_1, p(y_2) = x_2, \dots, p(y_s) = x_s$. Let's say $p(x) = k_0 + k_1 x + k_2 x^2 + \dots + k_{s-1} x^{s-1}$, we have

$$\begin{cases} k_0 + k_1 y_1 + k_2 y_1^2 + \dots + k_{s-1} y_1^{s-1} = x_1 \\ k_0 + k_1 y_2 + k_2 y_2^2 + \dots + k_{s-1} y_2^{s-1} = x_2 \\ \vdots \\ k_0 + k_1 y_s + k_2 y_s^2 + \dots + k_{s-1} y_s^{s-1} = x_s \end{cases}$$

The system of equation is equivalent to

$$\begin{bmatrix} 1 & y_1 & y_1^2 & \dots & y_1^{s-1} \\ 1 & y_2 & y_2^2 & \dots & y_2^{s-1} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & y_s & y_s^2 & \dots & y_s^{s-1} \end{bmatrix} \begin{bmatrix} k_0 \\ k_1 \\ k_2 \\ \vdots \\ k_{s-1} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_s \end{bmatrix}$$

Note that the matrix is a Vandermonde matrix. It's invertible if and only if $y_1, y_2, \dots, y_s$ are all distinct. So the matrix is invertible $\implies$ the system of equations has a unique solution $\implies p(x)$ exists. Therefore, there exists a degree $s - 1$ polynomial $p(x)$ such that $p(x_1^r) = x_1, p(x_2^r) = x_2, \dots, p(x_s^r) = x_s$.

Lemma proof ends.

Since A is positive definite, A has spectral decomposition $A = Q \Lambda Q^T$, where $\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$. Then

$A^r = Q \Lambda^r Q^T$. Let's say among all the eigenvalues, s of them are distinct ($s \leq n$). According to Lemma 1, there exists a polynomial $p(x)$ such that,

$$p(A^r) = Q p(\Lambda^r) Q^T = Q \begin{bmatrix} p(\lambda_1^r) & & & \\ & p(\lambda_2^r) & & \\ & & \ddots & \\ & & & p(\lambda_n^r) \end{bmatrix} Q^T = Q \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} Q^T = A.$$

It's obvious to see that if $XY = YX$, then $f(X)Y = Yf(X)$, where $f(x)$ is an arbitrary polynomial. Hence,

$$A^r B = B A^r \implies p(A^r) B = B p(A^r) \implies AB = BA.$$

9. True.

Proof:

Note that

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} = \begin{bmatrix} I & \\ B^T A^{-1} & I \end{bmatrix} \begin{bmatrix} A & \\ & C - B^T A^{-1} B \end{bmatrix} \begin{bmatrix} I & A^{-1} B \\ & I \end{bmatrix}$$

where $\begin{bmatrix} I & \\ B^T A^{-1} & I \end{bmatrix}$ is a unit lower triangular matrix and $\begin{bmatrix} I & A^{-1} B \\ & I \end{bmatrix} = \begin{bmatrix} I & \\ B^T A^{-1} & I \end{bmatrix}^T$. So $\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} = L \begin{bmatrix} A & \\ & C - B^T A^{-1} B \end{bmatrix}$

Hence, $S = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$ is positive definite $\iff S = LDL^T$ and the diagonal entries in D are positive $\iff$

$\exists L_1, L_2, A = L_1 D_1 L_1^T, C - B^T A^{-1} B = L_2 D_2 L_2^T$ where the diagonal entries in D_1, D_2 are positive and L_1, L_2 are unit lower triangular matrices $\iff A, C - B^T A^{-1} B$ are positive definite.

Problem 14.8.

Solution 14.8.

Set $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$, then $\frac{ax^2+bxxy+cy^2}{x^2+y^2} = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$, where $A = \begin{bmatrix} a & \frac{1}{2}b \\ \frac{1}{2}b & c \end{bmatrix}$. By Rayleigh Quotient,

$$\lambda_{\min} \leq \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \leq \lambda_{\max}$$

Therefore,

1. if $a = 0, b = 2, c = 0$, $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $\lambda_1 = 1, \lambda_2 = -1$. So maximum value is 1, minimum value is -1 .
2. if $a = 3, b = 2, c = 3$, $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, then $\lambda_1 = 4, \lambda_2 = 2$. So maximum value is 4, minimum value is 2.
3. if $a = 1, b = 2, c = 4$, $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $\lambda_1 = \frac{5+\sqrt{13}}{2}, \lambda_2 = \frac{5-\sqrt{13}}{2}$. So maximum value is $\frac{5+\sqrt{13}}{2}$, minimum value is $\frac{5-\sqrt{13}}{2}$.

Problem 14.9.

Solution 14.9.

1. $A = \begin{bmatrix} 3 & 4 & 0 \end{bmatrix}$, then $AA^T = \begin{bmatrix} 25 \end{bmatrix}$. The Spectral decomposition of AA^T is $Q\Lambda Q^T = \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} 25 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}$. So $U = \begin{bmatrix} 1 \end{bmatrix}, \Sigma = \begin{bmatrix} 5 & 0 & 0 \end{bmatrix}$. We have obtained U and Σ , let's calculate V now. The first column of V , $\mathbf{v}_1 = \frac{1}{\sigma} A^T U = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$,

then we extend $\mathbf{v}_1$ into ONB of $\mathbb{R}^3$, hence $\mathbf{v}_2 = \frac{1}{5} \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, so $V = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The SVD of A is

$$\begin{bmatrix} 3 & 4 & 0 \end{bmatrix} = U\Sigma V^T = \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}^T.$$

2. Similar to 1,

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = U\Sigma V^T = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{3} & 0 & -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \end{bmatrix}^T$$

3. By observation,

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 3 & & & \\ & 1 & & \\ & & 4 & \\ & & & 2 \end{bmatrix}$$

Obviously, $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ is orthogonal matrix because columns are ONB. So

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & & & \\ & 1 & & \\ & & 4 & \\ & & & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & & & \\ & 1 & & \\ & & 4 & \\ & & & 2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} = U\Sigma V^T$$

4. By observation,

$$\begin{bmatrix} A & O \\ O & O \end{bmatrix} = \begin{bmatrix} U\Sigma V^T & \\ & O \end{bmatrix} = \begin{bmatrix} U & \\ & I \end{bmatrix} \begin{bmatrix} \Sigma & \\ & O \end{bmatrix} \begin{bmatrix} V & \\ & I \end{bmatrix}^T$$

Problem 14.10.**Solution 14.10.**

Proof:

Let $\mathbf{v}$ be any eigenvectors with length 1, and λ is the corresponding eigenvalue. By the definition of operator norm,

$$\sigma_{\max} = \|A\| = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| \geq \|A\mathbf{v}\| = \|\lambda\mathbf{v}\| = |\lambda|\|\mathbf{v}\| = |\lambda|$$

So the absolute values of all eigenvalues are at most the largest singular value of A .

Problem 14.11.**Solution 14.11.**

Please read Example 8.12.19 before doing this problem.

First, center all the points so that the average of all $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Let $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ be the centered points, and $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} x-2 \\ y-9 \end{bmatrix}$.

The new points are $\begin{bmatrix} -2 \\ -9 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 11 \end{bmatrix}$. Then we form these vectors into a matrix and find the rank-one approximation of this matrix by the NumPy package in python.

```
>>> import numpy as np
>>> A = np.array([[ -2, 1, 1, 2], [ -9, -1, -1, 11]])
>>> U, Sigma, VT=np.linalg.svd(A)
>>> Sigma[0]*U[:, :1]@(VT[:, :1].T)
array([[ -1.71032454,  -0.47330423,  -0.32086529,  -2.00568613],
       [ -9.05471613,  -2.505744   ,  -1.69870926, -10.61840497]])
```

Now we get the rank-1 approximation of the matrix, and obviously we have

$$9.05471x_0 - 1.71032y_0 = 0 \implies 9.05471(x-2) - 1.71032(y-9) = 0 \implies 9.05471x - 1.71032y - 2.71654 = 0$$

Problem 14.12.**Solution 14.12.**

$$1. \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix}.$$

Set $A = \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix}$, and it is an upper triangular matrix, so the eigenvalues are the diagonal entries, that is,

$$\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0.$$

But $AA^T = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 0 \end{bmatrix}$, so the singular values are $\sigma_1 = \sigma_2 = \sigma_3 = 1, \sigma_4 = 0$.

$$2. \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ 10^{-4} & & & 0 \end{bmatrix}.$$

Set $A = \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ 10^{-4} & & & 0 \end{bmatrix}$, From Problem 13.4.4, we have $\lambda_1 = 10^{-1}, \lambda_2 = -1 \times 10^{-1}, \lambda_3 = i \times 10^{-1}, \lambda_4 = -i \times 10^{-1}$.

But $AA^T = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 10^{-8} \end{bmatrix}$, so the singular values are $\sigma_1 = \sigma_2 = \sigma_3 = 1, \sigma_4 = 10^{-4}$.